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MMPC 015 : RESEARCH METHODOLOGY FOR MANAGEMENT DECISIONS

This course consisting of four blocks spread over 14 units. **Block 1** on Introduction to research Methodology consists of three units.

Unit 1 discusses the meaning of research and the importance of research in decision-making. It also discusses the broader aspects of research methodology, types of research and the role of research in management decision-making. **Unit 2** discusses the steps involved in the business research process and defines the research problems. It also explains hypothesis testing, preparing the research design, collecting the data, analysis of data and preparation of the report or the thesis. **Unit 3** on Research Designs covered different types of research designs, with reference to exploratory, descriptive, and experimental research designs. The key considerations for adopting a specific research design are discussed. The lab and field experiments were discussed in detecting the cause-and-effect relationships. Further, Issues of internal and external validity were discussed. Also, decisions on choosing lab experiments or field experiments were examined.

Block 2 on Data Collection and Measurement consists of four units. **Unit 4** discusses the necessity and usefulness of data collection. It also distinguishes between primary and secondary data. A few methods of collecting primary data with their merits and demerits are explained. Designing and presenting a questionnaire for the collection of primary data find a place in this unit. The unit also explains how to go about editing primary data. **Unit 5** on attitude Measurement and Scales explains the type of managerial research problems which utilise the tools of attitude measurement. The unit describes the various issues in attitude measurement and explains various types of scales, namely the Nominal Scale, the Ordinal Scale, and the Interval Scale. Some commonly used scales in research like the Guttman Scale, Thurstone's equal appearing interval scale, the Semantic differential scale, the Likert scale, the Q-sort technique, and Multidimensional scaling also find a place in this unit. **Unit 6** discusses how to design a questionnaire for improved results properly. Questionnaire sequencing is critical for eliciting required information from participants. **Unit 7** on Sampling and Sampling Designs defines the various sampling concepts and explains the advantages of sampling over complete enumeration. It also describes various probability and non-probability sampling methods.

Block 3 on Data Presentation and Analysis consists of four units. **Unit 8** explains the significance of data presentation and describes various types of data classification. A few data tabulation and presentation devices are also developed in this unit. The subject matter of **Unit 9** is Nonparametric tests. These are also called distribution-free tests. In these tests, the researcher does not need to have knowledge of the distribution of the population parameter being tested. This unit explains the difference between parametric and nonparametric tests. The relevance of and various steps involved in using nonparametric tests are also discussed here. The unit concludes by explaining, with the help of appropriate examples, how to conduct one sample, two samples and more than two samples nonparametric tests. **Unit 10** on Multivariate Analysis of Data explains various multivariate techniques like multiple regression, discriminant analysis and factor

analysis for the analysis of relevant data for a research problem. The emphasis is on interpreting the findings of multivariate analysis in any research study. The unit also explains the use of a particular technique suitable for a particular research problem.

Block 4 on Report Writing and Presentation consists of four units. **Unit 11** on Ethics in Research has provided detailed information regarding research ethics and the principles involved in research ethics. The researcher should follow the research ethics so that their research should be unique. The researcher needs to see that it should be free from plagiarism and that the idea is not copied. Proper citations and references must be added whenever necessary in the research manuscript. **Unit 12** on Substance of Reports explains various types of reports. The unit also discusses how to prepare a proposal for a report and review the draft of a report. **Unit 13** deals with formats of reports. It distinguishes between and explains various parts of a report. The unit also gives guidelines for typing a report and for its editing by using copyreading and proofreading symbols. **Unit 14** on presentation of a report discusses the various components of presentation skills and also lists out the suggestions for preparation of AV aids, the usefulness of AV aids, and the care a presenter has to take while presenting a report.



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BLOCK 1

**INTRODUCTION TO
RESEARCH METHODOLOGY**

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UNIT 1 RESEARCH METHODOLOGY: AN OVERVIEW

Objectives

After going through this unit, you should be able to:

- Understand the meaning and importance of research
- Identify the need for research in decision making.
- Understand the role and purpose of research in important areas in business.

Structure

- 1.1 Introduction
- 1.2 Meaning of Research
- 1.3 Research Methodology
- 1.4 Research Method
- 1.5 Business Research Method
- 1.6 Types of Research
- 1.7 Importance of business research
- 1.8 Role of research in important areas
- 1.9 Summary
- 1.10 Self-assessment Exercises
- 1.11 Further Readings

1.1 INTRODUCTION

When managers use research, they are applying the methods of science to the art of management. All business undertakings operate in the world of uncertainty. There is no unique method which can entirely eliminate uncertainty. But research methodology, more than any other procedure, can minimise the degree of uncertainty. Thus, it reduces the probability of making a wrong choice amongst alternative courses of action. This is particularly significant in the light of increasing competition and growing size which make the task of choosing the best course of action difficult for any business enterprise.

It is imperative that any type of organisation in the present environment needs systematic supply of information coupled with tools of analysis for making sound decisions which involve minimum risk. It is in this context that the research methodology plays a very important role. In this unit, we will discuss at length the importance of research in decision making by delineating all its relevant elements.

1.2 MEANING OF RESEARCH

Research is not a fishing expedition or an encyclopaedic gathering of assorted facts. Research is purposeful investigation. The scientific method carries out a

detailed and comprehensive investigation of a given topic, concern, or issue. It provides a structure for decision making. In the real-world situation, need to make decisions to solve problems like social and environmental issues, increasing sales, employee retention etc. You must have a problem, worry, or issue that has transformed into a question before you can begin investigating it. There are three parts involved in any investigation: (1) the implicit question posed. (2) the explicit answer proposed. (3) collection, analysis, and interpretation of the information leading from the question to the answer. This third part is the defence that justifies the recommendation and is viewed as research. For example, consider the statement “We recommend that model A TV be priced at 14000/-”. This was the recommendation forwarded to the marketing vice president by the marketing research manager. The implicit question posed in this quote is what should be the selling price of model A? The explicit answer is Rs.14000/-. The third part deals with the collection, analysis, and interpretation of the information leading from the question to the answer of Rs.14000/-.

The research questions can be gleaned via observation of the outside world, past study, professional literature, or peer recommendations. Because your query must be answered, research should begin with the appropriate question. So, research is described as a methodical examination of a specific subject or problem using scientific methodologies. Business research generates and provides the needed qualitative or quantitative information to make business judgments, including problem-solving and decision-making actions.

The word research identifies a process by which the organisation attempts to supply the information required for making sound management decisions. Research is not synonymous with common sense. The difference revolves around words such as “systematic,” “objective,” and “reproducible”. Both research and common sense depend on information; the distinction between them lies in the procedures and methods adopted by which the information is obtained and used in arriving at conclusions. Research cannot address itself to the complete information on a particular subject. Hence two secondary characteristics of research specify “relevance”, and “control.”

A systematic approach is essential in good research. Each step must be so planned that it leads to the next step. It is usually very difficult to go back and correct the mistakes of the previous step; sometimes it is impossible. Even when it is possible, it will involve loss in time and money. Authors have divided research into a number of steps. Both the number of steps and the names are some what arbitrary; however, the recognition of a sequence is crucial. Planning and organisation are part of this systematic approach with a lot of emphasis given to the interdependence of the various steps.

While planning, one of the very common mistakes that committed is the separation of data collection and data analysis. First, we collect the data; then, we decide what analysis is appropriate. This approach invites a disaster. In one of the research projects, depth interviews of the fresh college students were carried out at a very high cost and the necessary data were all collected. The data were still unanalysed because no one knew how to proceed. Our point is that considerable thought should have been given at the planning stage itself as to what kind of analysis will be required for the project which will satisfy the needs of the decision maker.

Objectivity warrants an approach which is independent of the researcher's personal views and opinions with regard to the answer to the problem under investigation. It is possible to have honest differences with respect to the proper definition or collection procedure, but the one selected must not be chosen to verify a prior position.

Look at a scene in the morning and then in the evening. Use the naked eye and then the tinted glasses. It is the same with research. A high proportion of shoppers in store A have a positive opinion of store A. Shoppers in store B may have a totally different opinion of store A. Purchase behaviour varies with price specials. It is always possible to prove a point if one desires, by carefully selecting the respondents, time, and place. True research attempts to find out an unbiased answer to the decision-making problem.

A reproducible research procedure is one which an equally competent researcher could duplicate and from it obtain approximately the same results. In order to achieve reproducibility, all procedures must be stated unambiguously. Precise wording of questions, method of sampling, collection method, interviewer instructions, and all other details must be clearly stated. Even if the environment changes, the research is at least "conceptually" reproducible in the sense that the steps could be mentally duplicated.

The interviewer should avoid the temptation of rephrasing the question for the respondent in order to preserve reproducibility aspect. Poor and vague sampling procedure can also lead to non-reproducibility. If procedures are vague and not stated clearly, you cannot expect consistency even from the same interviewers.

Relevancy accomplishes two important tasks. First it avoids the collection of unnecessary information along with the accompanying cost. In the second place it forces the comparison of the data collected with the decision maker's criteria for action. Before the start of the research project, you should ask the question "what action would you take if the research answer were?" This approach enables both the investigator and the decision maker to know whether the project is on the right direction.

Control aspect is particularly elusive in research. We must be aware that the results of our study are due to the presence of some factor other than those we are investigating. It is impossible to have control on all other factors; the best we can do is to have control for those we think are most likely to cause us difficulty. Suppose we study the relationship between shopping behaviour and income without controlling for education and age, it will be the height of folly since our findings may reflect the effect of education or age rather than income.

Control raises extremely difficult issues when research is conducted in a live environment. Many factors other than the ones of principal interest may influence the research results. The danger is that the researcher may attribute changes to one variable when the uncontrolled variables are the causes.

Control must consider two aspects. (1) Those variables that are truly with in your control must be varied according to the nature of your investigation. (2) Those variables beyond your control should be recorded.

Activity 1

The three parts concerning any research investigation are:

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Activity 2

The five distinguishing features of any good research are:

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Activity 3

Mention a few research studies where it is impossible to have control on all other factors:

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1.3 RESEARCH METHODOLOGY

The term ‘Methodology’ is frequently used in research projects, and it gives a complete overview of the rationality of the research. Research Methodology refers how scientific research is carried out. It involves rationally adopting several methods to tackle research challenges in a systematic manner. Methodology aids in comprehending not just the results of scientific investigation, but also the method itself. In a broader way, methodology entails searching/linking methods and studying specific theories used in research field areas, designing the best strategy according to research goals and objectives.

Practically say ‘research methodology’ answer “how” of any piece of research is related to the research technique and how a researcher plans a study in a structured way to ensure reliable outcomes that address the study’s goals and objectives. Apart from that, research methodology should explain the design decisions by illustrating that the approaches and techniques chosen match the study objectives and goals and provide valid and accurate results. A suitable research methodology yields scientifically solid results, whereas a lousy methodology yields none. Hence, it should highlight some basic (What, How, Why) questions of our research. Such as:

1. What was the purpose of your research?
2. What types of ‘research method’ can we use and why?
3. What types of data should we consider for your research analysis purpose?

4. What were the data collecting methods?
5. How did you analyze the collected data?
6. What kind of resources has been used in your research?

In addition, methodologies encompass all approaches, strategies, and instruments used by a researcher to finish the experiment and solve the study challenge. While methods display the entire study framework, explaining its various research design components. After all, 'methodology' enables researchers to verify a study's accuracy and helps them identify the research method's credibility, validity, and dependability. Another way, we can say that methodology will justify choosing the proper methods. On the other side, 'research methods' give detailed information on the research design, participants, materials, equipment, variables, and processes.

1.4 RESEARCH METHOD

The tactics, procedures, or techniques used in collecting data or evidence for analysis to reveal new knowledge or generate a better understanding of a topic are referred to as research methods. The term "research methods" refers to all of the researcher's processes undergoing when investigating his study issue. As a result, techniques will be regarded as the heart of research methodology. There are several research procedures, each of which employs a unique set of data-gathering instruments. So, when preparing your approaches, you will need to make critical judgments over different methods used in the research. For instance, how do you collect data (Qualitative vs. quantitative) and how do you analyse the data (Descriptive vs. experimental). Thus, research methodology isn't simply about the research methods, but it is to consider the logic behind using the methods. Researchers need to identify the logic and rationality of using one approach over another.

Activity 4

Identify any three research methods

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Activity 5

Differentiate Methodology and Method

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1.5 BUSINESS RESEARCH METHOD

A business research method refers to a set of research procedures, tactics or techniques used by organizations to assess if a particular business venture is

worthwhile of their time and effort. Furthermore, research methodologies assist businesses in determining the feasibility of their firm and recommend the right marketing strategy for their offerings.

Business research may assist in determining the potential desires of the clients and allowing to design better goods and services. The nature of the business also draws attention to the unique difficulties that a company addresses. It highlights the major emphasis of the company's products and incorporates everything a firm undertakes to achieve its objectives.

Some of the critical areas where business research is used:-

- Reducing Risks. Any company, large or small, should strive to minimize risks wherever feasible.
- Identify threats for business.
- Helps business to build customer centric.
- Forecast revenue.
- Acquire a more competitive business.

1.6 TYPES OF RESEARCH

However, research definition also varies from expert to expert and domine to domine. In a larger context, variations of research definition could be understood and underpin depending on researchers' research philosophical thinking. The belief of particular philosophy shapes researchers' opinion to choose particular methods for finding their research question answer. In 'social science' domain perspectives, research can be classified into three broad categories (application, objective, and enquiry) based on their perspectives in the 'social science' domain.

When looking at a research project from the standpoint of its application, there are two major categories to consider: pure and applied research. Pure research deals with very abstract or specialized concepts, where the process involves examination, development, verification of research methods, or developing and testing the theories.

At the same time, research is classified as descriptive, correlation, and explanatory research based on their objectivity to do research. Descriptive research mainly describes a particular situation, individual, group, or phenomenon. This method is suitable when the specific research topic is less explored or proper knowledge is not in existence. The correlational base study mainly emphasizes finding an association between two or more aspects of a phenomenon. Rather than that, sometimes researchers attempt to do research for more clarification purposes, specifically to understand/explain (why, how) two or more aspects of a phenomenon, known as explanatory base research.

From the 'Mode of enquiry' perspective, social research mainly follows three broad approaches to find an answer related to the research question. One is a qualitative/unstructured research approach that gives researchers flexibility in all aspects of the research process; follows an unstructured, open, flexible way of enquiry. Study results mainly focus on its diversity, and narrative perspective

rather than an analytical or quantified perspective and primarily concentrate on the description of experience and feeling instead of their measurements. Another is the structured/quantitative approach, which is deeply guided by rationalism philosophy. The quantitative approach follows a proper predetermined, rigid, and structured set of methods aiming to measure the phenomenal variation of variables by giving more importance to the reliability and validity of findings and communicating these findings more collectively so that inferences can be generalized. However, researchers will choose or combine multiple methods for data collection and data analysis perspective for better research objectivity or problem representation purpose.

1.7 IMPORTANCE OF BUSINESS RESEARCH

Business Research plays a critical role in day-to-day business activities. It helps to solve simple to complex problems like increasing sales of a particular product in a particular market segment and introducing a new product or starting a new business. The below points are highlighting the important ways the research is important for businesses.

For new ideas and insights: Research inculcates scientific and rational thinking, and it promotes the development of logical habits of thinking and organization. The role of research in several fields of applied economics, whether related to business or the economy, has considerably increased in modern times. Research may mean the outlet for new ideas and insights to philosophers and thinkers.

Solving operational problems: The complex nature of business and government has focused attention on the use of research in solving operational problems. Research provides the basis for nearly all government policies in our economic systems.

Decisions for policymaking: Through research, we can devise alternative policies and examine the consequences of each policymaker. Decision-making may not be a part of research, but research certainly facilitates the policy maker's decisions.

Solving problems: Research has its special significance in solving business and industry operational and planning problems. Research is equally essential for social scientists in studying social relationships, seeking answers to various social problems, and providing practical guidance in solving immediate problems of human relations.

Firms also conducted research in their core areas to find problem-related specific solutions.

- a) **Products:** For testing/introducing new products for existing/new product-line extensions purposed. Identifying product core value attributes that attract consumers most, such as - Product packaging-related research and increased product shelf-life.
- b) **Price:** Understanding the target segments people's price-sensitiveness related good/service incurred cost. Understand whether consumers treated price as a significant entity for perceived value measurement.

- c) **Place:** Identification of suitable platform/areas for business transaction purposes and took a preventive measurement basis of satisfaction/corporation prospect between supplier and distributors.
- d) **Promotions:** Identify effective/efficient media platforms for running product ads and choosing telecast frequency-time ads. Besides that, decide how much money we should invest for promotional activity.

Business decisions: Operational research, market research, and motivational research are considered crucial, and their results assist in making business decisions in more than one way.

For students: To those students who are to write a master's or Ph.D. thesis, research may mean careerism or a way to attain a high position in the social structure. It helps to understand the "researcher" as a user of the library.

To generalize the new theories: To analysts and intellectuals, research may mean the generalizations of new theories. Develops a critical and scientific attitude, disciplined thinking or absent of mind' to observe objectively (scientific deduction & inductive thinking); research skills will pay off in the long term, particularly in the "age of information".

1.8 ROLE OF RESEARCH IN IMPORTANT AREAS

The purpose of research is to improve society and businesses by progressing knowledge through scientific theories, concepts, and ideas. Through research, an executive can quickly get a synopsis of the current scenario which improves his information base for making sound decisions affecting future operations of the enterprise.

Marketing

Marketing research has become very crucial in taking sound marketing decisions. Marketing research involves the process of systematic collection, compilation, analysis, and interpretation of relevant data for marketing decisions. Research tools are applied effectively for studies involving demand forecasting, consumer buying behaviour, measuring advertising effectiveness, media selection, test marketing, product positioning, and new product potential.

Production

Research enables an organisation to decide on what to produce, how much to produce, when to produce, and for whom to produce in the field of production. Research tools are also of immense help in quality control and setting up optimum inventory level.

Banking

Banking institutions have found it useful to setup research departments for the purpose of gathering and analysing information both for their internal operations and for making in-depth studies on economic conditions of business. Reserve Bank of India has setup an excellent research department for planning and management reporting.

The materials department uses research to frame suitable policies regarding where to buy, how much to buy, when to buy, and at what price to buy.

Human resource development

The human resource development department uses research to study wage rates, incentive schemes, cost of living, employee turnover rates, employment trends, and performance appraisal. It also uses research effectively for its most important activity namely manpower planning.

Government

Research lays the foundation for all government policies in our economic system. For example, research is applied for evolving the union finance budget and railway budget every year. Research is used for economic planning and optimum utilisation of resources for the development of the nation. Research is also needed for systematic collection of information on the economic and social structure of the nation. Such information indicates what is happening to the economy and what changes are taking place.

Activity 4

List out the uses of research in the field of:

- a) Hospital Management

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- b) Railways

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- c) Temple Management

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- d) Traffic Control (by Police)

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1.9 SUMMARY

We have started the discussion by emphasizing that research is the application of science to the art of management. Research lays the structure for decision making. Research is not synonymous with common sense. Research is characterised by - systematic, objective, reproducible, relevance, and control. Research methodology refers to conducting research in a systematic manner using different research methods and explain the rationality behind using certain methods. Research

methodology minimises the degree of uncertainty involved in management decisions.

Business research methods refers to a set of research procedures, tactics or techniques used by organization. Research plays a crucial role in businesses to solve simple to complex problems. The importance and role of research in the areas of management has been briefly covered. The areas include marketing, production, banking, materials, human resource development, and government.

1.10 SELF-ASSESSMENT EXERCISES

- 1) Briefly explain the meaning and importance of each of the following in research.
 - a) Systematic
 - b) Objectivity
 - c) Control
 - d) Relevance
 - e) Reproducible
- 2) Identify one or two articles referred to the business management. Read the article carefully. To what extent do you believe that business and management research should always meet the requirement of systematic, objective, relevance? Give reasons for your answer.
- 3) Outline the features that can make business research distinctive from research in other disciplines.
- 4) Briefly explain the role of research in managerial planning and decisions.
- 5) Discuss the role of research in managing businesses.

1.11 FURTHER READINGS

Ranjit Kumar, “Research Methodology- A step-by-step Guide for Beginners”. Sage Publishes, New Delhi.

Boyd, Westfall, and Stasch, “Marketing Research Text and Cases”. All India Traveller Bookseller, New Delhi.

Brown, F.E. “Marketing Research, a structure for decision making”, Addison - Wesley Publishing Company.

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Stockton and Clark, “Introduction to Business and Economic Statistics” D.B. Taraporevala Sons and Co. Private Limited, Bombay.

UNIT 2 STEPS FOR RESEARCH PROCESS

Objectives

After going through this unit, you should be able to:

- Identify the steps in the research process.
- State research problems clearly and precisely.
- Formulate the research problem in terms of a hypothesis to be tested.
- Understand the role and importance of literature review in research
- Perform literature review to solve business problems in the organization.

Structure

2.1 Introduction

2.2 Research process

2.3 Define research problems

2.3.1 Research Problem

2.3.2 Importance of research problem

2.3.3 Points to Ponder on Research Problem

2.3.4 Units of Analysis

2.3.5 Sources of research topic

2.3.6 Consideration in selecting a research problem

2.3.7 Steps in formulating a research problem

2.3.8 Criteria of Selection

2.3.9 Characteristics of Good Research Questions:

2.4 Research Problem as Hypothesis Testing

2.5 Extensive literature review in research

2.6 Development of working hypothesis

2.7 Preparing the research design

2.8 Collecting the data

2.9 Analysis of data

2.10 Preparation of the report or the thesis

2.11 Summary

2.12 Self-Assessment exercises

2.13 Further Readings

2.1 INTRODUCTION

In today's complicated globe, every culture is confronted with major social, political, and economic issues. These issues need to be addressed systematically, practically, and rationally, and issues to be solved logically and scientifically. However, a sequential follow process necessitates the acquisition of new information. Research allows for the accumulation of knowledge and wisdom.

In other words, a research methodology is a systematic act of obtaining facts, examining, and interpreting facts to solve issues that society faces.

2.2 RESEARCH PROCESS

Broadly, the research process is a roadmap for a researcher, and it gives an overview of the researcher's research journey. Similarly, consider a traveller undertaking a journey; travellers must decide where they want to go? Which route they want to prefer is based on a known or unknown path, and they should decide their traveling mode before starting their journey. Similarly, researchers must have clarified two critical questions: First one, 'What research question's answer you are finding'? And after deciding on the research question, the second question comes: 'how do you find out the research question's answer'? Research methodology is the process of determining the answer to your research question. So, to obtain the research question's answer, you must go through a series of experimental stages in your research process. However, the sequence of steps may change or is not fixed in nature; based on the researcher's experiences, steps may change. Ultimately, the researcher chooses a diversity of methods/procedures for the research process.

Research process consists of series of actions or Steps necessary to effectively carry out research and the desired sequencing of these steps. The following section will try to give a proper glance of the entire process to familiarise you with the numerous activities you'll need to do to complete your research, giving you a sense of what the research trip entails.

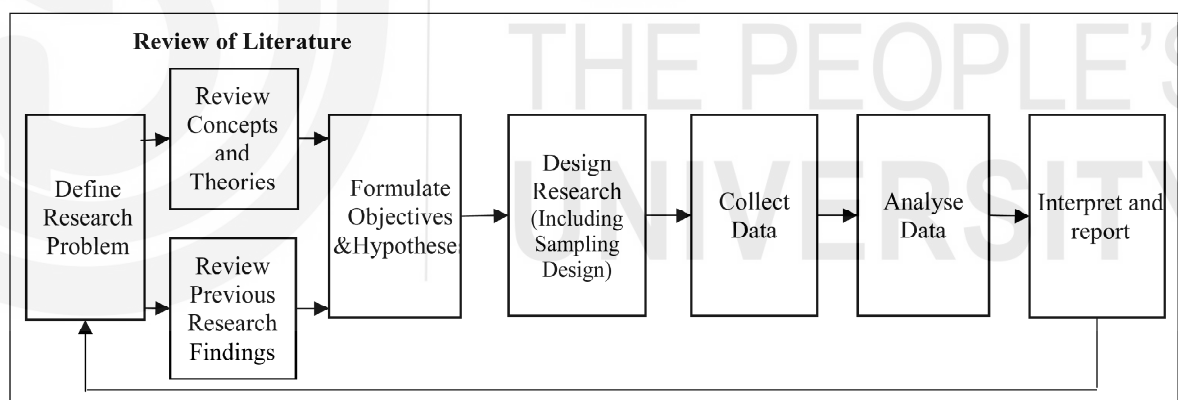


Figure 1 : Research Process

Figure 1 indicates the process consists of a number of closely related activities. If subsequent procedures have not been taken into account in the early stages, serious difficulties may arise which may even prevent the completion of the study.

2.3 DEFINE RESEARCH PROBLEMS

2.3.1 Research Problem

In general, any questions that researchers want to be answered or any assumption or assertion researchers wish to examine or investigate may be turned into a research topic or a study subject. However, keeping in mind that every question is not feasible to convert into a research problem because it might have difficulty

proving it. However, ‘formulating a research problem’ seems pretty easy for newcomers in a research area, but practically formulating a research problem in a meaningful way is not easy. Hence, it is advisable that researchers must spend reasonable time preparing their research problem.

A problem that someone would like to do research. It is anything that a person find unsatisfactory or unsettling, a difficulty of some sort, a state of affairs that need to be changed. A problem involve areas of concerns to researchers, for condition they want to improve, difficulties they want to eliminate, questions for which they want to seek answers. Research Problem refers to some difficulty/ need which a researcher experiences in the context of either theoretical or practical situation and wants to obtain a solution for the same. In simple words, research problem is a statement of the information needed by a decision maker to help solve a management decision problem.

2.3.2 Importance of research problem

Formulation of research problem is the first and most essential step in the research process, and it’s more like choosing your destination before starting the journey. Also, it’s impossible to conduct a good research study without an apparent research problem because it acts as a research foundation. Moreover, without a good research problem, researchers cannot determine its subsequence steps and cannot choose their methodology, directly affecting their economic plan.

2.3.3 Points to Ponder on Research Problem

The following points should be kept in mind while defining a research problem:

- i) The right question must be addressed if research is to aid decision makers. A correct answer to the wrong question leads either to poor advice or to no advice.
- ii) Very often in research problem we have a tendency to rationalize and defend our actions once we have embarked upon a particular research plan. The best time to review and consider alternative approaches is in the planning stage. If this is done needless cost of false start and redoing work could be avoided.
- iii) A good starting point in problem definition is to ask what the decision maker would like to know if the requested information could be obtained without error and without cost.
- iv) Another good rule to follow is “Never settle on a particular approach” without developing and considering at least one alternative”.
- v) The problem definition step of research is the determination and structuring of the decision maker’s question. It must be the decision maker’s question and not the researcher’s question.
- vi) What decision do you face? If you do not have decision to make, there is no research problem.
- vii) What are your alternatives? If there are no alternatives to choose, again there is no research problem.

- viii) What are your criteria for choosing the best alternative? If you do not have criteria for evaluation, again there is no research problem.
- ix) The researcher must avoid the acceptance of the superficial and the obvious.

Activity 1

Briefly mention advantages of considering nine points mentioned in 2.2 while defining a research problem.

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2.3.4 Units of Analysis

The individuals or objects whose characteristics are to be measured are called the units of analysis. They may be persons, groups of persons, business establishments, inanimate objects, transactions, monetary units, or just about objects or activity a person can name. Some very interesting communication studies have even used words as the units of analysis. Basically, the units answer the question, “What objects am I interested in?” Consideration of several alternatives for units will sharpen one’s thinking concerning the appropriate universe.

To illustrate the selection of units, consider a manufacturer of small electrical motors who wishes to ascertain the extent to which its potential customers know the company exists. The potential customers are basically business entities. But the units of the universe could also be defined as purchasing departments, production departments, engineering departments, or particular individuals within one or more departments. Again we come to all the pervasive question of what alternative actions are being considered by the manufacturer. In terms of these actions, who should be aware of the manufacturer’s existence? Is the company considering specific acts that might increase awareness levels for certain groups? These are the sorts of questions that should be considered in specifying the appropriate units of analysis.

Let us go a step further. Is each unit, however defined, equally important? Or does importance vary with the purchasing power of the potential customer? If purchasing power is the critical item, one procedure is to use the units as established by the prior thought pattern and to weight each by the purchasing power of the entity it represents. With this approach, no difficulty or complexity is introduced in the definition of the universe, but a complexity must be introduced later in the processing.

Alternatively, the basic unit of analysis could be defined in terms of transactions rather than in terms of potential buyers. With buyers as units, the universe consists of persons, groups of persons, or business entities. With transactions as units, the universe consists of activities as the focus of interest. Typically, in

research, we wish to classify or measure the units according to some characteristics. Once more we see the interdependence of research decisions: the selection of universe units is best determined only in conjunction with what is to be measured.

Is the manufacturer interested in finding the percentage of buyers who are aware of the company's existence? Or is the manufacturer more concerned with the - percentage of the transactions in the market place in which the buyer is aware (or unaware) of its existence? The same type of comparison would be required if level of awareness were measured; here it might be average level of awareness of buyers versus average level per transaction.

Rupee value would be still another basis for establishing units. These rupees could be rupees expended on small electrical motors of the type made by this manufacturer. The objective would then be to determine the percentage of the total rupee value market awareness of the company's existence. This was very too close to what the president had in mind. In theory the president wanted to classify every rupee spent as coming from a buyer who was or was not aware of the company's existence. The same arithmetic result is obtained if buyers are classified according to awareness with each buyer weighted by the rupee volume he or she generates.

The well-known management concept of a "decision-making unit" (DMU) often comes into play in defining the units of the universe. But the DMU is usually difficult to define in an unambiguous manner. A purchase that is a wife's decision in one family may be a husband's decision in another and a joint decision in third. How does one cope with this problem? A two-step procedure is a possibility. The first stage units are families; within each family the decision maker is identified. The units of the problem universe are the DMU's. Any compromise research universe must be evaluated against that concept, including the possibility that the DMU is a group. The following example from marketing will clarify the concepts.

The specification of the appropriate DMU for industrial products is more difficult than it is for consumer products. The number of persons who have potential involvement is greater. Job designations do not have the same meaning for all organisations. Responsibilities for ultimate decisions vary with size of organization, organizational structure, philosophy of decentralization, plus the personalities involved. The question is further complicated by the fact that some characteristics of interest refer to the organization-for example, size, geographic location, and past purchases-while others, such as preferences education, and attitudes, uniquely refer to specific individuals.

The problem definition, whether for a consumer product or an industrial product, must specify the units of analysis. It is better to err at this stage by specifying conceptually correct units that pose difficult problems in implementations. Compromises in the transition to operational definitions can then be better evaluated. This approach also permits the possibility of using different procedures with different market segments or a multistage approach in identifying the relevant DMU's.

Activity 2

Explain the meaning of DMU with an example.

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2.3.5 Sources of research topic

In general, humanities, occupational or academic base study revolves around four P's: problem, people, phenomena, and programs, and the emphasis of each 'P' varies base from study to study, but in general practices, research has considered at least two Ps. Every study has two aspects area; 'people' provides researcher's 'study population,' and the other three P's (problem, phenomena, and programs) provide 'subject area.'

Research must decide the general area of interest or aspect of a subject matter. Initially the problems may be stated in broad general way. However, understanding the problem thoroughly the problem statement should rephrase into meaningful terms from an analytical point of view. The best way of understanding the problem is to discuss with colleagues, experts, in colleges and educational institutes with guides for help to find the problem. Researcher must at the time examine all available literature to get himself acquainted with the selected problem.

2.3.6 Consideration in selecting a research problem

There are many important things to keep in mind while considering a research topic or problem. These considerable parameters will ultimately ensure their study goes smoothly, in a manageable way, so they do not lose their motivation.

- i) **Interest:** This is the primary criterion for selecting any topic or problem because the study journey entails many more difficult, time-consuming, and unforeseen problems. So, if the study problem does not capture the researchers' attention, it won't be easy to maintain the same level of motivation for the entire research time frame.
- ii) **Magnitude:** A researcher must have good knowledge about the research process. So that they can correctly visualize the task involved related to the proposed study.
- iii) **Measurement of concepts:** While using any concepts in your study, make its indicator and measurement clear. For example, an investigator wants to measure the effectiveness of a health promotional program, so before starting research work, they must be clear about what it determines and how they measure it.
- iv) **Level of expertise:** As a researcher, you have to ensure that you have proper knowledge or expertise for your proposed task.
- v) **Relevance:** As a researcher, make sure your research topic has relevance

in terms of your profession, gives value to the current value of knowledge, and acts as a bridge of current research gaps.

2.3.7 Steps in formulating a research problem

The following steps will help those researchers who does not even decided their research area, topic, and questions. The following process will help researchers narrow down their vast research area into subareas that could become your research problem.

- i) Identification of broad/subject areas according to your interest.
- ii) Separate the broad area into sub-areas.
- iii) Select one sub-topic or sub-areas that is perceived most in interest.
- iv) Raise a question and list out all possible questions that comes to mind and if the listed question is too many for a manageable perspective, then follow step i-iii.
- v) Formulate research objectives and sub-objectives, which depend on research questions.
- vi) Evaluate objective viability in light of human, time, and financial resources.

This task of formulating or defining, a research problem is a step of greatest importance in the entire research process. The above steps in formulating a research problem can be divided into two activities:

- i) Identification/Selection of the Research Problem
- ii) Formulation of the Research Problem

i) Identification / Selection of the Research Problem

This step involves identification of a few problems and selection of one out of them, after evaluating the alternatives against certain selection criteria. There are different source from which the researcher can find the research problem. The sources of research problems can be Reading, academic experience, daily experience, and exposure to field situations, consultations, brainstorming and intuition.

2.3.8 Criteria of Selection

The selection of one appropriate researchable problem out of the identified problems requires evaluation of those alternatives against certain criteria. They are:

- **Internal/Personal criteria** – Researcher's Interest, Researcher's Competence, Researcher's own Resource: finance and time.
- **External Criteria or Factors**– Researchability of the problem, Importance and Urgency, Novelty of the Problem, Feasibility, Facilities, Usefulness and Social Relevance, Research Personnel.

ii) Formulation of the Research Problem

Formulation is the process of refining the research ideas into research questions and objectives. Formulation means translating and transforming the selected research problem/topic/idea into a scientifically researchable question. It is concerned with specifying exactly what the research problem is. Essentially, two steps are involved in formulating research problems:

- a) ***Understanding the problem thoroughly.*** The most important way of understanding the problem is discussion with colleagues or persons who are expert in knowledge or skill. The problem is also studied using literature survey. After understanding the problem, it is expressed, in more specific terms to make the problem, clear.
- b) ***Expressing the problem in more specifies terms or, more clear.*** Problem definition or problem statement is a clear, precise and concise statement of the question or issue that is to be investigated with the goal of finding an answer or solution.

There are two ways of stating a problem:

- ✓ Posting question / questions
- ✓ Making declarative statement / statements

Hence, the steps involve in the process of defining the problem as below:

1. Statement of the problem in a general way
2. Understanding the nature of problem
3. Surveying the available literature
4. Developing ideas through discussions
5. Rephrasing the research problem

The good research problem should meet the criteria of clear and unambiguous, empirically possible to solve, should be able to verify with facts, interesting to pursue finding the solution, and should be able to check the availability of supervision.

2.3.9 Characteristics of Good Research Questions:

The good research questions must have below four important characteristics:

- a) The question is feasible (i.e., It can be investigated without an undue amount of time, energy, or money)
- b) The question is clear (i.e., most people would agree as to what the key words in the question mean)
- c) The question is significant (i.e., it is worth investigating it contribute important knowledge about human condition)
- d) The question is ethical (i.e., it will not involve physical or psychological harm or damage to human feelings, or to natural or social environment of which they are part)

Example 1:

Consider the following problem

“Why is the **Productivity** of **Kathmandu** Metropolis is higher than **Pokhara** Municipality?”

This problem is too **general** so this problem should be expressed in more **specific forms** as

“What factors are responsible for increasing the Kathmandu’s labour productivity in Five manufacturing industries in 2009 related to Pokhara’s corresponding figures?” This type of problem is specific, and the conclusion drawn through research is considered to be more reliable.

Example 2:

- **Management Problem**

- Placement office has noticed, while major companies make annual recruiting visits to campus for engineers, not many national or local companies are formally recruiting business majors through the placement office
- Why? How do we address this?

- **Research Problems**

- Why are companies not taking advantage of the resources that the placement service offers? Are companies going around the service?
- Are companies aware of the UAH placement service?
- Are companies aware of the reputation of the UAH Business School?
- What kind of things might generate more recruiting activity?

- **Marketing Research Objectives**

- To determine to what extent companies are aware of the UAH placement service
- Determine whether companies, especially locals, are aware of the strong reputation of the UAH Business School
- To determine whether a quarterly newsletter highlighting UAH business programs and students might generate more recruiting activity.

Example 3:

- **Management Problem**

- What price should we charge for our new product?

- **Research Problem**

- What are our costs of production and marketing (COGS)?
- What are our pricing objectives and position in the market?
- What price does similar type of products sell for?

- What is the perceived value of our product in the marketplace?
- Are there any norms or conventional practices in the marketplace (e.g., customary prices, continual discounting)
- **Research Objectives**
 - To assess the costs involved in producing and selling our product
 - To determine corporate objectives and their implications for pricing
 - To examine current prices for direct and indirect competition
 - To determine potential customer reaction to various prices and their perception of the benefits of owning the product

2.4 RESEARCH PROBLEM AS HYPOTHESIS TESTING

It is often convenient to structure a research problem in terms of a hypothesis to be tested. The hypothesis must be agreed upon by both the manager and the researcher, although the formal statement is primarily the responsibility of the researcher. In this use of the word, a hypothesis is simply a statement about the universe. It may or may not be true; the research is designed to ascertain the truth. Consider the following pair of hypotheses.

H0: At least 10 % of the viewing audience for “children’s” TV shows consists of adults.

H1: Less than 10 % of the viewing audience for “children’s” TV shows consists of adults.

First, it should be noted that these hypotheses are worded in such a way that either one or the other is correct. They cannot both be correct, and they cannot both be incorrect. Second, in order for these two hypotheses to be useful in a research decision making situation, the decision maker should choose first act if the first is true, and a second act if the second act is true. Both statements are characteristic of all situations in which a research problem is properly structured in terms of hypothesis testing.

The terminology “state of nature” is often used to refer to the true situation in the universe. For example, the advertising manager for a firm selling a product frequently purchased by adults is considering the possibility of advertising the product on children’s TV shows.

Table 1 shows a structuring of the hypothesis with respect to the decision making. In this case, the hypothesis has been constructed so that if H0 is true, the recommendation is to advertise while if H1 is true, the recommendation is not to advertise. This procedure can be extended to any number of alternatives or options. The basic rule is that each hypothesis under test would lead to a specific recommendation if it were true. If any alternatives enumerated would not be adopted regardless of the research findings, those alternatives can be eliminated without any research. If several hypotheses would lead to the same

recommendation, there is no need to identify which of these hypotheses true-all hypothesis is leading to the same alternative can be grouped together into a single hypothesis.

Table 1: States of Nature versus Recommended Decisions. Properly Structured Research Problem

State of nature	Recommended Decision	
	Advertise	Don't Advertise
H0 True	X	
H1 True		X

It has established the minimum purchasing power required for the medium and large stores as Rs.100 million and Rs.250 million, respectively. A possible structuring of the three hypothesis is

H0: Total purchasing power is less than Rs. 100 million,

H1: Total purchasing power is between Rs. 100 million and Rs. 250 million.

H2: Total purchasing power is greater than Rs. 250 million.

The acceptance of one of these hypotheses leads directly to one of the three actions contemplated: It is not necessary to determine purchasing power precisely than that indicated in the three hypotheses. A two-stage research project might be indicated. Stage one would establish whether purchasing power were clearly within the values specified by the hypotheses. Only if stage one revealed a figure close to Rs. 100 million or Rs. 250 million would stage two be undertaken. There is no reason to incur the cost of determining purchasing power with precision unless that precision is required in decision making.

Decision making as hypotheses testing is a two-step process with error possibilities at each step. At step one there is the relationship between the states. of nature and the action recommended. The percentage of adults in the audience may not be a proper guide to action. The number of adults may be better guide. Rupee value expenditure in the product category may be still better. At step two there is the possibility that the research result may be erroneous with respect to the state of nature. The sample may indicate that the percentage of adults in the audience is less than 10 %. Or the opposite error may occur. Research procedures do not yield certainty with respect to the true state of nature. No matter how careful we are, we may conclude that HO is true when H1 is true or vice versa. This fact means the decision maker and the researcher must evaluate the seriousness of different kinds of errors. The seriousness of the errors can be appraised only in terms of the actions that will be recommended. No harm occurs until the conclusions have an impact on what the firm does.

Table 2 presents the general situation with two possible states of nature and two alternative actions. Assuming the relationship between the two states of nature and the two action is valid, no error occurs if we conclude that HO is true and it is true; similarly, no errors occur if we conclude that H1 is true, and it is true. Errors occur in each of the other cells.

Table 2: True States of Nature versus Conclusions. Possible Types of Errors

State of nature	Recommended Decision	
	H0 True	H1 True
H0 True	No Error	Type 1 Error
H1 True	Type 2 Error	No Error

Consider the format of Table 3 in terms of the problem dealing with advertising on children's TV shows. H₀ (at least 10 % of the audience is composed of adults) leads to a recommendation to advertise. H₁ (less than 10% of the audience is composed of adults) leads to a recommendation not to advertise. Under the Type II error we think H₀ is true and recommend in favour of advertising, but less than 10% of the audience is composed of adults. The opposite situation exists with the Type 1 error. We recommend against advertising (thinking H₁ is true), but at least 10% of the audience is composed of adults. Which error is more serious-failing to take advantage of an existing opportunity or incurring expenditures? Placed in those terms, it should be obvious that no general answer can be given to the general question of which error is more serious. The decision maker must carefully evaluate each specific situation with its unique set of potential risks and benefits. The identification of which error is more serious is not a meaningless intellectual exercise! The analysis can be planned in such a way so as to minimise those errors that are viewed as more serious.

Activity 6

A hypothesis is universe

State of nature refers to universe

Activity 7

Consider any research problem of interest to you.. Structure it in terms of an appropriate hypothesis to be tested. Also mention the possible errors and recommendations associated with your hypothesis. Illustrate your answer with the help of a suitable example.

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2.5 EXTENSIVE LITERATURE REVIEW IN RESEARCH

Once the problem is formulated,a brief summary of it should be written down. The researcher should undertake extensive literature survey connected with the problem. For this purpose, the abstracting and indexing journals and published

or unpublished bibliographies are the first place to go to. Academic journals, conference proceedings, Govt. reports, books etc. must be tapped depending on the nature of the problem. A good library will be a great help to the researcher at this stage.

As a researcher, the essential task is to go through the available literature to acquire all possible knowledge related to your chosen topic area. However, reviewing the literature is time-consuming and frustrating, but its support role is vast, and it has a significant contribution in every research operational stage. The begging stage helps researchers clarify research ideas, establish deep theoretical roots, and develop solid methodological knowledge. After that, a literature review helps to consolidate and improve your research area knowledge and validate your finding by linking pre-established knowledge. Researcher may review two types of literature:

- i) The conceptual literature concerning the concepts and theories and
- ii) The empirical literature consisting of studies made earlier which are similar to the one proposed.

Below few points are presented in a summary form as its function.

1. It gives researchers a theoretical base for their study.
2. It established the path between suggested study and the previous study's findings.
3. It allows you to demonstrate how your discoveries contribute to your field's current body of knowledge.

2.5.1 Role of literature review on research

Gaps identification:

Literature review act as a paradox because it is necessary to have some basic idea of the problem want to study. Another hand, a literature review effectively influences the choice or puts conditions on thinking towards the research problem. The literature review clarifies the research topic by gaining more profound knowledge and further clarifies the research problem by creating a more precise or realistic research problem. Further, it helps researchers identify the research gap by understanding other investigators' research findings and what suggestions they have made.

Improving current research methodology:

Through literature review, researchers can choose appropriate methods for their current study. When researchers review the previous study similar to their problem field, they will know more about procedures and methods respectively understand each method's drawbacks. So, by becoming aware of each method's possible drawbacks and pitfalls, researchers can choose the best suitable methods that will give the most valid answer to your research question.

Positioning finding in an existing body of knowledge: -

Obtaining answers through research is a comparatively easy task relative to positioning your finding in the existing body of knowledge. Finding the research

question's answers must contribute to an existing body of knowledge by explaining its uniqueness from previous research findings and how it will fill the current knowledge gaps.

Sources of existing literature:

To adequately search for literature on your subject of interest, you must first understand the subject area and the problem you desire to examine to set search parameters. After that, build a bibliography for this vast topic. Researchers mainly refer journals, conference proceedings, Govt. reports, books etc.

2.6 DEVELOPMENT OF WORKING HYPOTHESIS:

Following a thorough literature review, the researcher should state the working hypothesis in plain language. The development of research hypotheses is especially significant because they serve as the focal point of study.

The formulation of working hypotheses is critical in most types of research. The hypothesis's job is to assist the researcher by limiting the scope of the inquiry and keeping them on track. It sharpens his thoughts and directs his attention to the more critical aspects of the situation. It also specifies the sort of data requested and the data analysis procedures to be employed. Developing working hypotheses by using the following approach.

- ✓ Discussions with colleagues and subject matter experts regarding the issue
- ✓ Examination of the data and records
- ✓ Examine similar studies in the area or studies on similar issues.

Working hypothesis should not be used, especially by formulative or exploratory who do not intend to test the hypothesis. However, as a general rule, the identification of working hypotheses is another fundamental phase in the research process.

2.7 PREPARING THE RESEARCH DESIGN:

The research problem having been formulated in clear cut terms, the researcher will be required to prepare a research design i.e., Researcher will have to state the conceptual structure with in which research would be conducted. "A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure". Different research designs can be conveniently described if we categorize them as:

- a) Research design in case of *exploratory research* studies
- b) Research design in case of *descriptive and diagnostic research* studies
- c) Research design in case of *Experimental research* design

The preparation of the research design, appropriate for a particular research problem, involves usually the consideration of the following:

- a. The means of obtaining the information.
- b. The availability and skills of the researcher and his staff.
- c. Explanation of the way in which selected means of obtaining information will be organized and the reasoning leading to the selection.
- d. The time availability for research.
- e. The cost factor relating to research i.e. the finance available for the purpose.

2.7.1 Sample design:

Sampling can be defined as the selection of some part of an aggregate or totality, the bases of which a judgment or inference about the aggregate or totality is made. A sampling is a small proportion of a population selected for observation and analysis. It is a collection of items or elements from a population.

Usually determining the sample design is a part of research design. Sample design refers to the target populations from which the data would be collected and how the data would be collected. All the items under consideration in any field of inquiry constitute a “universe” or “population”. A complete enumeration of all the items in the “population” is known as a “census inquiry”. Besides, this type of inquiry involves great deal of time, money and energy. Census enquiry is not possible in practice under many circumstances

Hence, quite often we select only a few items from the universe for our study purposes. For instance, blood testing is done only on sample basis. The items so selected constitute what is technically called a “sample.” The researcher must decide the way of selecting sample or what is popularly known as the “sample design”.

A sample design is a definite plan for obtaining a sample from a given population. It refers to the procedure the researcher will adopt in selecting items from the sample. It is designed before data collection. In designing a sample, the researcher must consider three things: sampling frame, selection of sampling items and sample size. The basic components of a sample design are:

- Choosing the sample units (who are to be surveyed)
- Choosing the sample size (how many to be surveyed)
- Choosing the sample procedure (how to ensure that those who are to be interviewed are included in the sample)
- Choosing the media (how to reach respondents in the sample-through mail survey, personal interview, telephone interview.)

The selected respondents constitute what is technically called a ‘sample’ and the method of selection is called ‘sampling technique.’ Researcher must prepare a sample design which decides how a sample should be selected and of what size such as sample would be.

Samples can be either probability or non-probability/non-random samples. Random sampling methods would be collected from the samples are those based on, simple random sampling, systematic sampling, stratified sampling, cluster/area sampling. Probability sampling is also known as 'random sampling' or 'chance sampling'. Under this sampling design, every item of the universe has an equal chance of inclusion in the sample. It is, so to say, a lottery method in which individual units are picked up from the whole group not deliberately but by some mechanical process.

Non – probability samples: samples are those based on, convenience sampling, and judgmental sampling quota sampling techniques. Nonprobability also called as purposive or deliberate sampling. Non-probability sampling is also known by different names such as deliberate sampling, purposive sampling, and judgment sampling. In this type of sampling, items for the sample are selected deliberately by the researcher; his choice concerning the items remains supreme.

2.8 COLLECTING THE DATA

Data collection is also known as field work. It involves administrating the research tools to gather data. It connects link to the reality of the work for the researchers. Data collection consists of taking ordered information from reality and transferring to some recording systems so that social behaviour can be understood and predicted. It is based on research design. Data collection completely fulfils the data requirements of a research project. It is the connecting link for the researchers to the world of reality. It provides the sources of comparative data by which data can be interpreted and evaluated against each other. Based on the data collection, data are presented and analysed. It suggests the type and method of data for meeting the information needed. It serves as a source of future reference and evidence because they are used to prepare written records and helps to takes ordered information from reality and transferring into some recording system so that it can be later examined and analysed.

In dealing with any real life problem it is often found data at hand are inadequate, and hence it becomes necessary to collect data that are appropriate. There are several ways of collecting the appropriate data which differ considerably in context of money costs, time and other resources at the disposal of the researcher.

Primary data can be collected either through experiment or through survey. If the researcher conducts an experiment, he observes some quantitative measures. "The researcher should select the methods of collecting the data taking into consideration the nature of investigation, objective and scope of the enquiry, financial resources, available time and the desired degree of accuracy".

2.9 ANALYSIS OF DATA:

The process of analysis begins after the data have been collected. After the data have been collected, the researcher turns to the task of analysing them. During the analysis stage several interrelated procedures are performed to summarize and rearrange the data. Researcher should classify the raw data into some purposeful and usable categories.

The goal of most research is to provide information. There is a difference between raw data and information. Information refers to a body of facts that are in a format suitable for decision making, whereas data are simply recorded measures of certain phenomena. The raw data collected in the field must be transformed into information that will answer the manager's questions.

The conversion of raw data into information requires that the data be edited and coded so that the data may be transferred to a computer or other data storage medium. If a database is large, there are many advantages to utilizing a computer. Assuming a large database, entering the data into the computer follows the coding procedure.

The analysis of data requires a number of closely related operations such as establishment of categories, the application of these categories to raw data through coding, editing, and tabulation, analysing and drawing inferences.

Coding operation is usually done at this stage through which the categories of data are transformed into symbols that may be tabulated and counted. Editing is the procedure that improves the quality of the data for coding. With coding the stage is ready for tabulation. Tabulation is a part of the technical procedure where in the classified data are put in the form of tables. Analysing work after tabulation is generally based on the computation of various percentages, co-efficient, etc..., by applying various well defined statistical formulas.

After analysing the data as stated above, the researcher is in a position to test the hypothesis, if any, he had formulated earlier. "Do the facts support the hypothesis or they happen to be contrary? This is the usual question which should be answered while testing hypothesis.

Various tests, such as chi square test, t-test, f-test, have been developed by statisticians for the purpose. The hypothesis may be tested through the use of one or more of such tests, depending upon the nature and object of research inquiry. If the researcher had no hypothesis to start with, generalizations established on the basis of data may be stated as hypothesis to be tested by subsequent researchers in time to come.

2.9.1 Generalization and interpretation:

If a hypothesis is tested and upheld several times, it may be possible for the researcher to arrive at generalization, i.e. to build a theory. As a matter of fact, the real value of researcher lies in its ability to arrive at certain generalizations. If the researcher had no hypothesis to start with, he might seek to explain his findings on the basis of some theory, it is known as interpretation. The process of interpretation may quite often triggers off new questions which in turn may lead to further research.

2.10 PREPARATION OF THE REPORT OR THE THESIS:

Finally, the researcher has to prepare the report of what has been done by him; writing of report must be done with great care keeping in view the following:

i) **The layout of the report should be as follow**

- a) **Preliminary pages:** In its preliminary pages the report should carry title and data followed by acknowledgements and foreword. Then there should be a table of contents followed by a list of tables and list of Graphs and charts, if any, given in the reports.

ii) **The main text** of the report should have the following parts;

- a) **Introduction:** it should contain the clear statement of the research and an explanation of the methodology adopted in accomplishing the research. The scope of the study along with various limitations should as well be stated in this part.
- b) **Main report:** the main body of the report should be presented in logical sequence and broken – down into readily identifiable sections.
- c) **Summary of findings:** after introduction there would appear a statement of findings and recommendations in non-technical language. If the findings are extensive, they should be summarized.
- d) **Conclusions:** towards the end of the main text, researcher should again putdown the results of the research clearly and precisely. In fact, it is the final summing up.

- iii) **At the end of the report,** appendices should be enlisted in respect of all technical data. Bibliography. i.e., list of books, journals, reports etc., consulted, should also be given in the end. Index should also be given specially in a published research report. Report should be written in a concise and objective style in simple language avoiding vague expressions such as “it seems”, “there may be “, and the like”. Charts and illustrations in the main report should be used only if they present the information more clearly and forcibly. Calculated “confidence limits” must be mentioned and the various constraints experienced in conducting research operations may as well be stated.

2.11 SUMMARY

The unit discussed the steps involved in the business research process. The research process involves the seven important steps - problem definition, review of literature, formulating objectives & hypotheses, research design, data collection, data analysis, and interpretation of results. All these steps have been explained in detail with their key elements. However, the unit explains the first two steps, namely formulating the research problem and literature review, more in detail. The following units discuss the remaining steps in detail.

The research problem identification is the first and most important step of the research process. The researcher must understand the sources and process of formulating the research problem. Both reviews of literature and research problem steps are complementary, as researcher need to perform review of literature to formulate the research problem and without knowledge gained through review of literature cannot solve the research problem.

2.12 SELF-ASSESSMENT EXERCISES

- a. How do you describe the research process?
- b. You are interested in doing some research on the interface between business organisations and schools. Write three research questions that may be appropriate.
- c. Give an original example of a problem in which the unit of analysis is a DMU (Decision-making-unit). Make sure to include a discussion of how to measure the principal characteristic of interest for DMU's which involve more than a single individual.
- d. "The problem definition stage is perhaps more critical in the research process than the problem solution stage." Discuss the statement.
- e. Below is the gist of an article from Business Week. After reading it,
 - i. Identify the broad problem area
 - ii. Define the problem

"While Chrysler's minivans, pickups, and sport utility vehicles take a big share of the truck market, its cars trail behind those of GM, Ford, Honda, and Toyota. Quality problems include, among other things, water leaks and defective parts."
- f. What is the problem statement in the following situation?

"Companies benefit through employee loyalty. Crude downsizing in organizations during the recession crushed the loyalty of millions. The economic benefits of loyalty embrace lower recruitment and training costs, higher productivity of workers, customer satisfaction, and the boost to morale of fresh recruits. In order that these benefits are not lost, some companies while downsizing try various gimmicks. Flex leave, for instance, is one. This helps employees receive 20% of their salary, plus employer provided benefits, while they take a 6 to 12 month sabbatical, with a call option on their services. Others try alternatives like more communication, hand holding, and the like."
- g. A sports goods company wishes to test two types of tennis rackets in order to determine which one is "best".
 - i. Propose and defend a precise definition of "best".
 - ii. What is the set of hypotheses that should be tested?
 - iii. What action would be associated with each hypothesis?

Why are critical reviews of relevant literature important in research studies?

- h. Describe the purpose of a literature survey.

2.13 FURTHER READINGS

Ranjit Kumar, “Research Methodology- A step-by-step Guide for Beginners”. Sage Publishes, New Delhi.

Boyd, Westfall, and Stasch, “Marketing Research Text and Cases”. All India Traveller Bookseller, New Delhi.

Uma Sekaran, Research Methods for Business – A skill building approach, Wiley Publishers.

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UNIT 3 RESEARCH DESIGNS

Objectives

After going through this unit, you should be able to:

- Discuss the various research designs as powerful tools to study the cause and effect relationships amongst variables in research.
- Explain the assumptions embodied in the design models.
- Choose the appropriate design model for a specific research problem.

Structure

- 3.1 Introduction
- 3.2 Functions and Goals of Research Design
- 3.3 Characteristics of a Good Design
- 3.4 Different Types of Research Designs
- 3.5 Exploratory Research Design
 - 3.5.1 Techniques used for exploration
 - 3.5.2 Characteristic of Exploratory research
- 3.6 Descriptive Research Design
- 3.7 Experimental Research Design
 - 3.7.1 Lab-Experiment
 - 3.7.2 The Field Experiment
- 3.8 Types of Experimental Designs
 - 3.8.1 Quasi and True Experimental Designs
 - 3.8.2 Simulation
 - 3.8.3 Decision on Experimental designs
- 3.9 Summary
- 3.10 Self-Assessment exercises
- 3.11 Further Readings

3.1 INTRODUCTION

The research design refers to a researcher's roadmap, which they decide to follow during their research journey to find their research question's answer and validate accurately, economically, and objectively as possible. The research design process allows the researcher and writers to follow the instruction of their research work. It tells the researcher what and how a distinctive method could be applied in different phases of the research. By making a research design plan, researchers are ready to decide and even ready to communicate with others or themselves about their decisions regarding the proposed study design. How they will collect information and select respondents, analyse the information, and communicate to other their findings.

Research design means an overall framework or plan for the activities to be undertaken during the course of a research study. When particular research area has been defined, research problem is defined, and the related literature in the area has been reviewed, the next step is to construct the research design. It is fundamental to the success of any scientific research. It involves decisions regarding what, where, when, how much and by what means concerning a research study. It constitutes a blue print for the collection, measurement and analysis of data. It serves as a framework for the study, guiding the collection and analysis of data, research instrument to be utilized and the sampling plan to be followed.

According to Kerlinger: *“Research design is the plan, structure and strategy of investigation conceived so as to obtain answers to research questions and to control variance.”*

According to Kinner and Taylor: *“A research design is the basic plan which guides the data collection and analysis phase of the research project. It is the framework which specifies the type of information to be collected, the source of data and the data collection procedure.”*

The formidable problem that follows the task of defining the research problem is the preparation of the design of the research project, popularly known as the “research design”. “A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure”. In fact, the research design is the conceptual structure within which research is conducted; it constitutes the blueprint for the collection, measurement and analysis of data. As such the design includes an outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data. More explicitly, the design decisions happen to be in respect of:

- i) What is the study about?
- ii) Why is the study being made?
- iii) Where will the study be carried out?
- iv) What type of data is required?
- v) Where can the required data are found?
- vi) What periods of time will the study include?
- vii) What will be the sample design?
- viii) What techniques of data collection will be used?
- ix) How will the data be analysed?
- x) In what style will the report be prepared?

3.2 FUNCTIONS AND GOALS OF RESEARCH DESIGN

Research design is needed because it facilitates the smooth sailing of the various research operations, thereby making research as efficient as possible yielding maximal information with minimal expenditure of effort, time and money. Just as for better, economical and attractive construction of a house, we need a blueprint

(or what is commonly called the map of the house) well thought out and prepared by an expert architect, similarly we need a research design or a plan in advance of data collection and analysis for our research project.

Research design stands for advance planning of the methods to be adopted for collecting the relevant data and the techniques to be used in their analysis, keeping in view the objective of the research and the availability of staff, time and money. Preparation of the research design should be done with great care as any error in it may upset the entire project. Research design, in fact, has a great bearing on the reliability of the results arrived at and as such constitutes the firm foundation of the entire edifice of the research work.

In brief, research design must, at least, contain:

- (a) A clear statement of the research problem;
- (b) Procedures and techniques to be used for gathering information;
- (c) The population to be studied; and
- (d) Methods to be used in processing and analyzing data.

3.3 CHARACTERISTICS OF A GOOD DESIGN

From what has been stated above, we can state the important features of a research design as under:

- It is a plan that specifies the sources and types of information relevant to the research problem.
- It is a strategy specifying which approach will be used for gathering and analyzing the data.
- It also includes the time and cost budgets since most studies are done under these two constraints.

3.4 DIFFERENT TYPES OF RESEARCH DESIGNS

Different research design can be conveniently described if we categorize them as:

- a) Research design in case of *exploratory research* studies
- b) Research design in case of *descriptive and diagnostic research* studies
- c) Research design in case of *Experimental research* design

Activity 1

List out the characteristic of a good research design

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3.5 EXPLORATORY RESEARCH DESIGN

Exploratory research studies are also termed as formulative research studies. The main purpose of such studies is that of formulating a problem for more precise investigation or of developing the working hypothesis from an operational point of view. The major emphasis in such studies is on the discovery of ideas and insights. Exploration is suitable in cases where researchers have no clear idea about upcoming problems they have met during the study.

Through this research, researchers will develop or clarify core study concepts, establish priorities, and formulate operational definitions that will help them improve their final research design. Sometimes researchers would like to do some exploration related to the study areas because the study area might be vague or new. They need to learn/identify the corresponding areas, new dilemmas, and new essential variables. Apart from that, researchers doing exploratory studies related to the proposed study area might know the practical feasibility. Despite its larger scope, still researchers and managers give less importance due to higher time taken, non-representative, subjectiveness, non-systematic design. As such the research design appropriate for such studies must be flexible enough to provide opportunity for considering different aspects of a problem under study. Inbuilt flexibility in research design is needed because the research problem, broadly defined initially, is transformed into one with more precise meaning in exploratory studies, which fact may necessitate changes in the research procedure for gathering relevant data.

3.5.1 Techniques used for exploration

While researching something with little information may appear challenging, various approaches may assist a researcher in determining the optimal study design, data gathering methods, and subject selection. There are two methods for conducting research: primary and secondary. A researcher can employ a variety of methodologies under these two categories. Apart from that, two techniques (qualitative and quantitative) are applicable for data gathering, but in adaptable prospects, researchers will commonly follow the qualitative techniques. Once researchers know the scope of qualitative research, they can easily adapt various approaches for exploratory investigation. Some approaches are: -

- Researchers use depth interview techniques, mainly in an unstructured way.
- Participant observation (what participants will do under the specific study setting)
- Researchers capturing (audio, video, picture format) the group living lifestyle considered current study.
- Case study techniques are used to understand a few conditions or events in-depth.
- Documents analysis
- Expert interview is performed to retrieve important or in-depth study subject's information.

3.5.2 Characteristic of Exploratory research

- Exploratory research base study is commonly unstructured, open-ended, high-time consuming and interactive in nature because research's have to collect all available information related to particular study.
- Researchers do the study commonly to know the answer of What contain bases question (Example, what is the purposed/problem of the study?).
- Research will do exploratory research specific to proposed study area, when there is a less existence of prior research as well as existing study not enough to answer the problem precisely.

Activity 2

Give three examples of exploratory research designs

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3.6 DESCRIPTIVE RESEARCH DESIGN

Descriptive research studies are those studies which are concerned with describing the characteristics of a particular individual, or of a group. Most of the social research comes under this category. Studies concerned with specific predictions, with narration of facts and characteristics concerning individual, group or situation are all examples of descriptive research studies. The researcher must be able to define clearly, what he wants to measure and must find adequate methods for measuring it along with a clear cut definition of 'population' he wants to study.

Since the aim is to obtain complete and accurate information, the procedure to be used must be carefully planned in this said study. The research design must make enough provision for protection against bias and must maximize reliability, with due concern for the economical completion of the research study. The design in such studies must be rigid and not flexible and must focus attention on the following:

- (a) Formulating the objective of the study (what the study is about and why is it being made?)
- (b) Designing the methods of data collection (what techniques of gathering data will be adopted?)
- (c) Selecting the sample (how much material will be needed?)
- (d) Collecting the data (where can the required data be found and with what time period should the data be related?)
- (e) Processing and analyzing the data.
- (f) Reporting the findings.

The difference between research designs in respect of the above two types of research studies can be conveniently summarized in tabular form as under:

	Exploratory Research Design	Descriptive Research Design
Overall design	Flexible design (design must provide opportunity for considering different aspects of the problem)	Rigid design (design must make enough provision for protection against bias and must maximize reliability)
Sampling design	Non-probability design (purposive or judgment sampling)	Probability sampling design (Random sampling)
Statistical design	No pre-planned design for analysis	Pre-planned design for analysis
Observational (data collection) design	Unstructured instruments for collection of data	Structured or well thought out instruments for collection of data

Activity 3

Distinguish between exploratory and descriptive research designs

3.7 EXPERIMENTAL RESEARCH DESIGN

Experimental studies are those where the researcher tests the hypotheses of causal relationships between variables. Such studies require procedures that will not only reduce bias and increase reliability, but will permit drawing inferences about causality. Usually experiments meet this requirement. Hence, when we talk of research design in such studies, we often mean the design of experiments. Today, the experimental designs are being used in researches relating to phenomena of several disciplines.

In the form of research the marketer tries to determine if the manipulation of the variable, called the independent variable, affects another variable, called the dependent variable. If casual research is performed well marketers may be able to use results for forecasting what might happen if the changes are made.

Experiments are much more effective than descriptive techniques in establishing the casual relationships. First, the units to be studied are selected by the researcher and each unit is assigned to the group determined by the researcher. The units do not select their groups, thus avoiding the self-selection bias. Second, a necessary consequence of the first, the researcher administers the predetermined treatment or treatments to the units within each group.

The use of a control group is almost mandatory in experimental designs. The inclusion of a control group permits a better isolation of the treatment component through a proper design like a simple cross sectional design.

A major contribution that the statisticians have made to experimental design is the development of randomization concept which enables the researcher to reduce the effect of the uncontrolled variables on comparative measures of response to the variables that are under the experimenter's control. Randomization is a useful device for ensuring on the average, that uncontrolled variables do not favour one treatment versus others.

To establish a causal-effects relationship or to measure one variable's effects (independent variables) on other variables (dependent variables), we need to control other factors/variables because it might be possible that other factors also influence dependent variables. However, it might be hard to control other covariates factors that may directly/indirectly manipulate the casual relation, especially in dynamic environmental settings due to its non-static or naturally flow events characteristic. Hence, experimental design commonly conducted in two setting contexts (a) in artificial setting where experimental are conducted in tight contrived environment also known as "Lab-Experiment". (b) In a natural setting experimental are conducted where activities occur on a regular basis, also known as "Field Experiment".

3.7.1 Lab-Experiment

When research particularly interest to observe the effects of independent on dependent variable or clearly measure the cause-effects relation between two variables (Independent, dependent), that time they controlled the other variables which might be possible to contained the study relation. Apart from that, some extent independent variables manipulation also needs in order to determine the degree of its causal effects. Laboratory setting is the best for manipulating and controlling specifically to test the variables causal-effects. Hence, we use 'Control' and 'Manipulation' terms.

- **Control**

Supposed say, we postulate the two variables (X, Y) 'causal-and-effects' relation. Where X and Y respectively treated as independent and dependent variables and it is feasible that another variable, say A, will also impact the dependent variable (Y). In that scenario, finding Y's actual occur variation due to X is quite not possible because we do not know the A variables actual effects on X. For example, in web page creation context, a firm's HR manager decide to give special training for new haired employee. Specifically, to prove his/her VP (Vice-president) that training programme would improve newly employee performance. However, it's quite possible that some of new employed performance are effective compared to other due to previous experience. So, in order to find the truth programme outcome, programme conductor must control the learner previous experience by excluding those new employees who had previous experience in web page creation.

- **Manipulation of the Independent Variable**

In order to examine the causal effects of an independent variable on a dependent variable, certain manipulations need to be tried. Manipulation simply means that

we create different levels of the independent variable to assess the impact on the dependent variable. For example, we may want to test the theory that depth of knowledge of various manufacturing technologies is caused by rotating the employees on all the jobs on the production line and in the design department, over a 4-week period. Then we can manipulate the independent variable, “rotation of employees,” by rotating one group of production workers and exposing them to all the systems during the 4-week period, rotating another group of workers only partially during the 4 weeks (i.e., exposing them to only half of the manufacturing technologies), and leaving the third group to continue to do what they are currently doing, without any special rotation. By measuring the depth of knowledge of these groups both before and after the manipulation (also known as the “treatment”), it would be possible to assess the extent to which the treatment caused the effect, after controlling the contaminating factors. If deep knowledge is indeed caused by rotation and exposure, the results would show that the third group had the lowest increase in depth of knowledge, the second group had some significant increase, and the first group had the greatest gains!

3.7.2 The Field Experiment

A field experiment, as the name implies, is an experiment done in the natural environment in which work goes on as usual, but treatments are given to one or more groups. Thus in the field experiment, even though it may not be possible to control all the nuisance variables because members cannot be either randomly assigned to groups, or matched, the treatment can still be manipulated. Control groups could also be set up in the field experiments. The experimental and control groups in the field experiment could be made up of the people working at several plants within a certain radius, or from the different shifts in the same plant, or in some other way. If there are three different shifts in a production plant, for instance, and the effects of the piece-rate system are to be studied, one of the shifts can be used as the control group, and the two other shifts given two different treatments or the same treatment—that is, different piece rates or the same piece rate. Any cause-and-effect relationship found under these conditions would have wider generalizability to other similar production settings, even though we may not be sure to what extent the piece rates alone were the cause of the increase in productivity, because some of the other confounding variables could not be controlled.

- **Internal Validity and External Validity**

Internal validity refers to the confidence we place in the cause-and-effect relationship. In other words, it addresses the question, “To what extent does the research design permit us to say that the independent variable A causes a change in the dependent variable B?” In research with high internal validity, we are relatively better able to argue that the relationship is causal, whereas in studies with low internal validity, causality cannot be inferred at all. In lab experiments where cause-and-effect relationships are substantiated, internal validity can be said to be high.

External validity refers to the extent of generalizability of the results of a causal study to other settings, people, or events, and internal validity refers to the degree of our confidence in the causal effects (i.e., that variable X cause variable Y). To what extent would the results found in the lab setting be transferable or

generalizable to the actual organizational or field settings? In other words, if we do find a cause-and-effect relationship after conducting a lab experiment, can we then confidently say that the same cause-and-effect relationship will also hold true in the organizational setting?

Field experiments have more external validity (i.e., the results are more generalizable to other similar organizational settings), but less internal validity (i.e., we cannot be certain of the extent to which variable X alone causes variable Y). Note that in the lab experiment, the reverse is true. The internal validity is high but the external validity is rather low.

Activity 4

Define the terms control and manipulation.

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3.8 TYPES OF EXPERIMENTAL DESIGNS

The different types of experimental designs and the extent to which internal validity is met in each are discussed in this section.

3.8.1 Quasi and True Experimental Designs

Some studies expose an experimental group to a treatment and measure its effects. Such an experimental design is the weakest of all designs, and it does not measure the true cause-and-effect relationship. This is so because there is no comparison between groups, nor any recording of the status of the dependent variable as it was prior to the experimental treatment and how it changed after the treatment. In the absence of such control, the study is of no scientific value in determining cause-and-effect relationships. Hence, such a design is referred to as a quasi-experimental design. The true-experimental designs, which include both the treatment and control groups and record information both before and after the experimental group is exposed to the treatment, are known as ex post facto experimental designs.

3.8.2 Simulation

An alternative to lab and field experimentation currently being used in business research is simulation. Simulation uses a model-building technique to determine the effects of changes, and computer-based simulations are becoming popular in business research. A simulation can be thought of as an experiment conducted in a specially created setting that very closely represents the natural environment in which activities are usually carried on. In that sense, the simulation lies somewhere between a lab and a field experiment, insofar as the environment is artificially created but not far different from “reality.” Participants are exposed to real-world experiences over a period of time, lasting anywhere from several hours to several weeks, and they can be randomly assigned to different treatment

groups. If managerial behaviour as a function of a specific treatment is to be studied, subjects will be asked to operate in an environment very much like an office, with desks, chairs, cabinets, telephones, and the like. Members will be randomly assigned the roles of directors, managers, clerks, and so on, and specific stimuli will be presented to them. Thus, while the researcher would retain control over the assignment and manipulation, the subjects would be left free to operate as in a real office. In essence, some factors will be built into or incorporated in the simulated system and others left free to vary (participants' behaviour, within the rules of the game). Data on the dependent variable can be obtained through observation, videotaping, audio recording, interviews, or questionnaires.

3.8.3 Decision on Experimental designs

Before using experimental designs in research studies, it is essential to consider whether they are necessary at all, and if so, at what level of sophistication. This is because experimental designs call for special efforts and varying degrees of interference with the natural flow of activities. Some questions that need to be addressed in making these decisions, are the following:

- Is it really necessary to identify causal relationships, or would it suffice if the correlates that account for the variance in the dependent variable were known?
- If it is important to trace the causal relationships, which of the two, internal validity or external validity, is needed more, or are both needed? If only internal validity is important, a carefully designed lab experiment would be the answer; if generalizability is the more important criterion, then a field experiment would be called for; if both are equally important, then a lab study should be first undertaken, followed by a field experiment, if the results of the former warrant the latter.
- Is cost an important factor in the study? If so, would a less rather than more sophisticated experimental design do?

These decision points are illustrated in the chart in Figure 3.1. Though managers may not often be interested in cause-and-effect relationships, a good knowledge of experimental designs could foster some pilot studies to be undertaken to examine if factors such as bonus systems, piece rates, rest pauses, and so on lead to positive outcomes such as better motivation, improved job performance, and other favourable working conditions at the workplace. Marketing managers would be able to use experimental designs to study the effects on sales of advertisements, sales promotions, pricing, and the like. Awareness of the usefulness of simulation as a research tool can also result in creative research endeavours in the management area, as it currently does in the manufacturing side of businesses.

3.9 SUMMARY

This unit covered different types of research designs, with particular reference to exploratory, descriptive and experimental research designs. The key considerations for adopting a specific research design is discussed. The lab and field experiments were discussed in detecting the cause-and-effect relationships. Further, Issues of internal and external validity were discussed.

Also, decision on choosing the lab experiments or field experiments were examined.

3.10 SELF-ASSESSMENT EXERCISES

- a) A foreman thinks that the low efficiency of the machine tool operators is directly linked to the high level of fumes emitted in the workshop. He would like to prove this to his supervisor through a research study.
- a) Would this be a causal or a correlational study? Why?
- b) Is this an exploratory, descriptive, or hypothesis-testing study? Why?
- c) What kind of a study would this be: field study, lab experiment, or field experiment? Why?
- b) Dr. Larry Norton of Memorial Sloan-Kettering Cancer Center predicts that cancer treatment will undergo major changes. Several drugs are being developed to battle cancer without harming healthy tissues. It is a question of discovering which of these drugs does the job best.

Design a study that would help find which drug would do the trick.

- c) Is a field study totally out of the question if one is trying to establish cause-and-effect relationships?
- d) In what ways do lab experiments differ from field experiments?
- e) Explain the concept of “trade-off between internal validity and external validity.”
- f) Comment on the following statement: “Because the external validity of lab experiments is not usually high, they are useless for investigating cause and effect relationships in organizations.”

3.11 FURTHER READINGS

Ranjit Kumar, “Research Methodology- A step-by-step Guide for Beginners”. Sage Publishes, New Delhi.

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